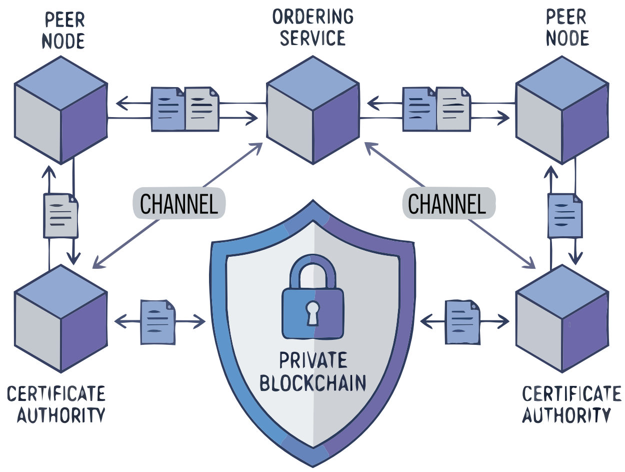
Hyperledger Fabric: Building a Private Blockchain

Hyperledger Fabric is a blockchain framework used in production by companies like Walmart and Maersk. This tutorial will help you set up a complete Fabric network on a Windows 11 laptop using Docker and WSL 2.



Public blockchains like Bitcoin let anyone join them, and every transaction is visible to all. In a private blockchain, however, the identities of the people joining the network are known. Think of a public blockchain as an open park where anyone can walk in and see what's happening. A private blockchain is more like a members-only club where only approved people can gain access. Unlike public chains, there is no mining, no cryptocurrency, and no anonymous participation in a private blockchain.

Most business use cases today require participants to identify and verify themselves. This is where Hyperledger Fabric proves useful. Maintained by the Linux Foundation, it uses smart contracts which are self-executing programs stored on the blockchain. These run automatically when certain conditions are met. They ensure quick and verified outcomes without a central authority.

Every participant in this blockchain holds a cryptographic identity issued by a CA (Certificate Authority). Business logic runs inside containers called chaincode, which is Fabric's version of smart contracts. Transactions go through an ordering service before being committed to the ledger, which is a record of all transaction logs.

Now let us build a private blockchain network using the official test network from the Fabric samples repository *(https://github.com/hyperledger/fabric-samples)*. It runs eight Docker containers that simulate two peer organisations, an ordering service, and three Certificate Authorities all on one desktop.

Table 1 lists every container the test network brings up, its role, and the port it listens on.

Table 1: Hyperledger Fabric test-network components

Container	Role	Image	Port
peer0.org1	Endorses and commits transactions for Org1	hyperledger/fabric-peer:2.5	7051
peer0.org2	Endorses and commits transactions for Org2	hyperledger/fabric-peer:2.5	9051
orderer	Orders transactions into blocks	hyperledger/fabric-orderer:2.5	7050
ca_org1	Issues X.509 identities for Org1	hyperledger/fabric-ca:1.5	7054
ca_org2	Issues X.509 identities for Org2	hyperledger/fabric-ca:1.5	9054
ca_orderer	Issues identity for the ordering service	hyperledger/fabric-ca:1.5	11054
couchdb0	State database for peer0.org1	couchdb:3.3.3	5984
couchdb1	State database for peer0.org2	couchdb:3.3.3	7984

When a transaction is initiated, it goes to both peer nodes for authorisation. Each peer runs the chaincode and signs the result. These signed approvals are then passed to the orderer, which lines transactions in sequence and bundles them together into blocks. The block is then sent